

	Work Instruction	Doc. Revision	00
	AXI V810 Y-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

This procedure is applicable to the following models:

Machine Model	Server Option		
	S1	S2	S3
V810 / V810i STD			
V810 / V810i XXL			
V810 / V810i S2EX		√	√
V810 / V810i Mini			
V810 / V810i XLT / XLTM / XLL		√	√
V810 / V810i XLW			
V810 / V810i S3		√	√

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	30 APR 2024	First Release	Yeap Leong Wei

	Work Instruction	Doc. Revision	00
	AXI V810 Y-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

Content

Content.....	2
Scope	3
Tool List	3
Origin Y-Axis Encoder Head Cable Removal Procedure	5
New Y-Axis Read Head Encoder Replacement	11
Tool Removal and Cleaning	15
Reference for Alignment Procedure	15

	Work Instruction	Doc. Revision	00
	AXI V810 Y-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

Scope

This procedure provides guidelines for the removal and replacing the Y-Axis Quantic Encoder Read Head.

Tool List

No	Part No	Description	Qty	UOM	Images
1	-	Ball Point Metric Allen Wrench Set	1	SET	
2	-	Metric T-Handle Wrench (M5 size)	1	PC	
3	-	Camera Tool	1	PC	
4	-	Tamper Proof Screw Driver Set	1	Set	
5	-	Cutter	1	PC	

	Work Instruction	Doc. Revision	00
	AXI V810 Y-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

6	4XASPP-A440	Renishaw Quantic Encoder Alignment Kit	1	Set	
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Table 1: Tool List

	Work Instruction	Doc. Revision	00
	AXI V810 Y-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

Origin Y-Axis Encoder Head Cable Removal Procedure

1. **Board Unloading:** Prior to commencing the removal procedure, ensure that any boards within the machine are safely unloaded. Follow the manufacturer's guidelines or equipment manual for proper board removal procedures.
2. **Shutdown Sequence:** Close V810 GUI and initiate the shutdown sequence for the Super Micro PC by following the manufacturer's recommended procedure. Ensure all data is saved and any necessary backups are performed before proceeding.
3. **Deactivation of Equipment:** At the operator console panel, press the “X-Ray Off” button to deactivate any X-ray equipment. Turn the key to the “Off” position to ensure complete deactivation of the system and associated components.
4. **Power Down Procedure:** Safely power down the SSCA main power source by following the manufacturer's recommended shutdown procedure. Ensure all power indicators are off before proceeding with the removal process.
5. **Access Door Opening:** Open all front access doors carefully to provide sufficient access to the internal components of the machine. Ensure doors are fully opened and secured to prevent accidental closure during the removal process.
6. **Left Outer Barrier Removal:** Remove the left outer barrier by unscrewing the door screws using the appropriate tools. Store the screws in a secure location for reassembly after the removal process is complete.

a) For S2EX & XLT



Figure 1: Left Outer Barrier (S2EX & XLT)

	Work Instruction	Doc. Revision	00
	AXI V810 Y-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

b) For S3



Figure 2:Left Outer Barrier (S3)

7. **Rear Access Door Opening:** Open the rear access door carefully, ensuring there are no cables or obstructions that may hinder the removal process. Take caution to prevent strain on any cables or connectors located near the access door.
8. **Removal of the vertical lead cover:** Unfasten the screws and remove the vertical lead cover. Take caution to prevent damage to the lead cover by placing it in flat and cleaned surface.

a) For S2EX, XLT

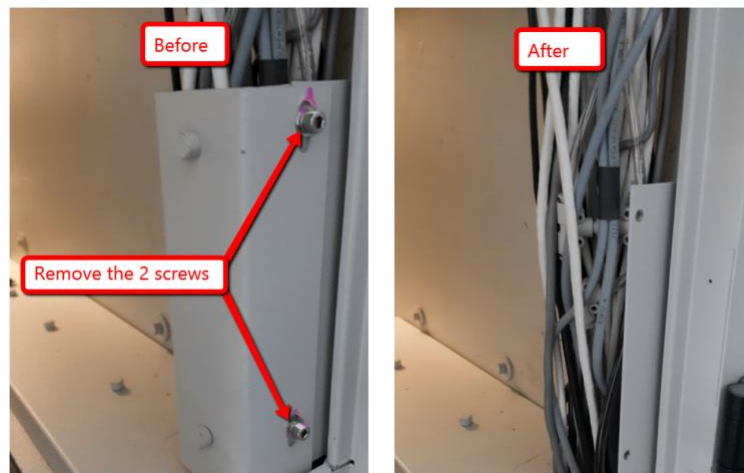


Figure 3:Vertical Lead Cover (Sample Image)

	Work Instruction	Doc. Revision	00
	AXI V810 Y-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

b) For S3

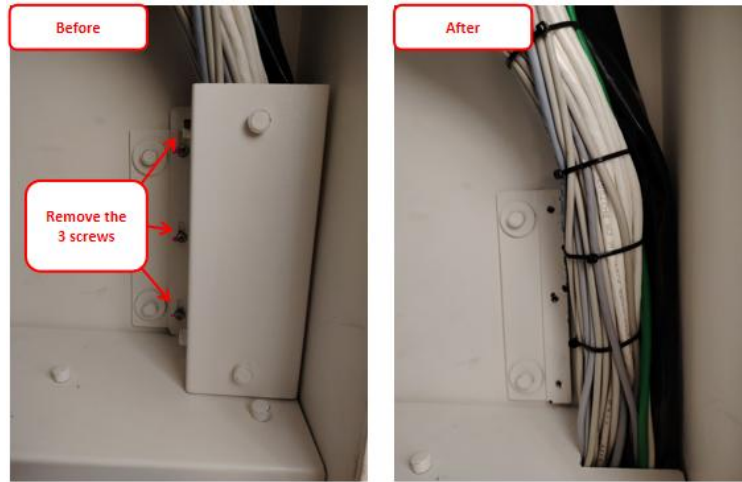


Figure 4: Vertical Lead Cover

9. **Removal of the horizontal lead cover:** Unfasten 3 screws and remove the horizontal lead cover. Take caution to prevent damage to the lead cover by placing it in flat and cleaned surface.

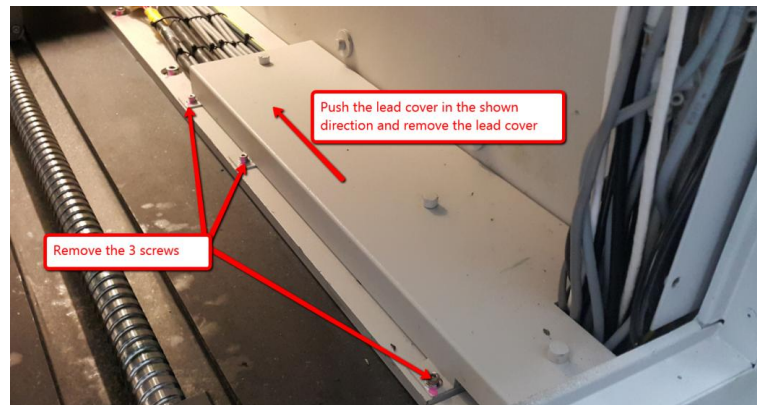


Figure 5: Horizontal Lead Cover (Sample Image)

10. **Removal of the SSCA Cable Bracket or Cable Tray Cover:** Remove the bracket or cable tray cover for the facilitation of the cable removal process.

a) For S2EX and XLT

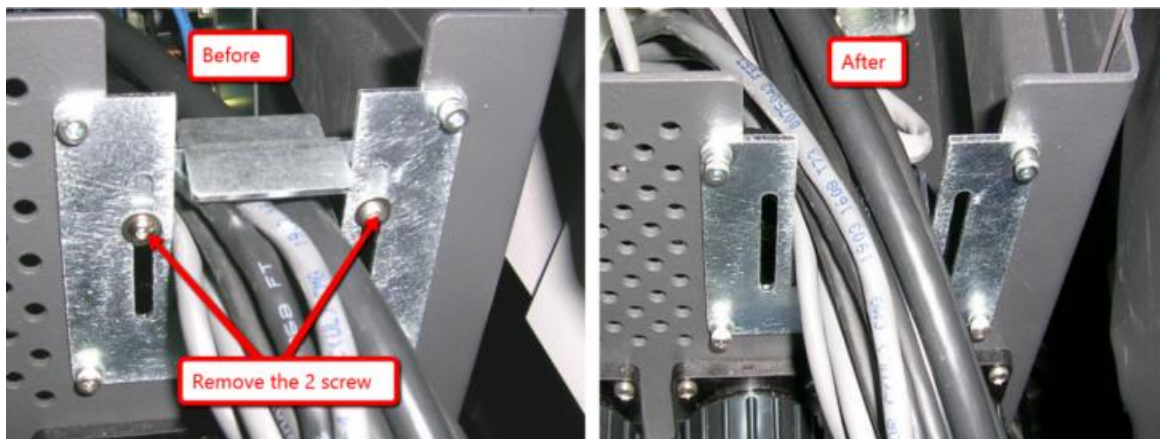


Figure 6: SSCA (Sample Image)

	Work Instruction	Doc. Revision	00
	AXI V810 Y-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

b) For S3

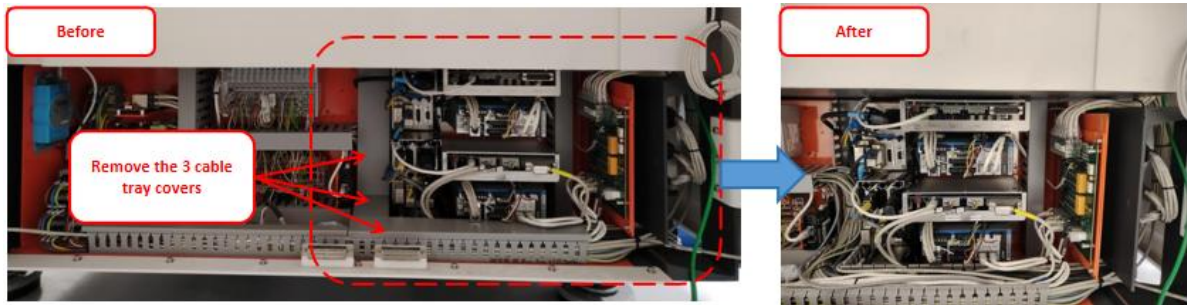


Figure 7:SSCA for S3 (Sample Image)

11. **Disconnected the Y-Axis Encoder Head Connector:** Unplug Y-Axis read head encoder connector from Y-Axis Kollmorgen kit (for S2EX) or from the extension cable connector (for XLT and S3).

Note: Remove necessary cable ties and carefully remove the encoder cable.

a) For S2EX and XLT

Note: For XLT, please unplug encoder read head from the extension cable connector.

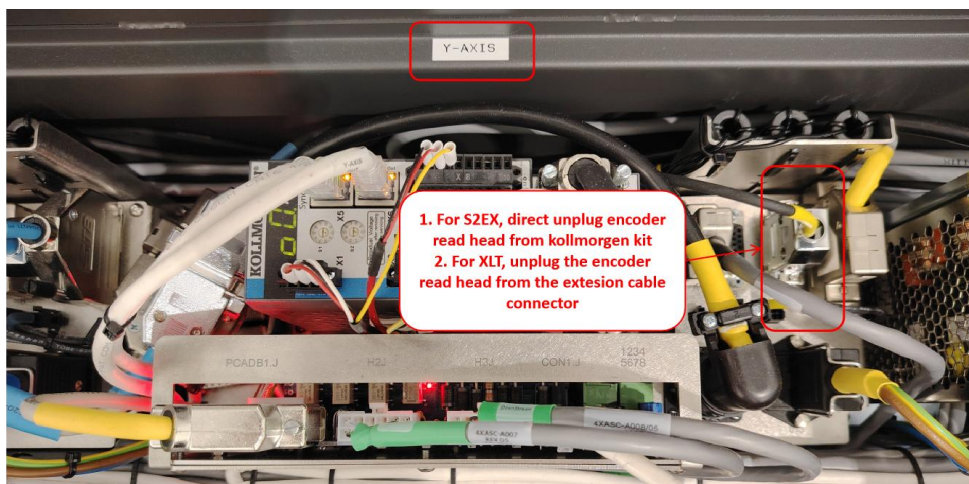


Figure 8:Y-Axis Kollmorgen Kit

b) For S3

Note: Please unplug the encoder read head from extension cable connector.

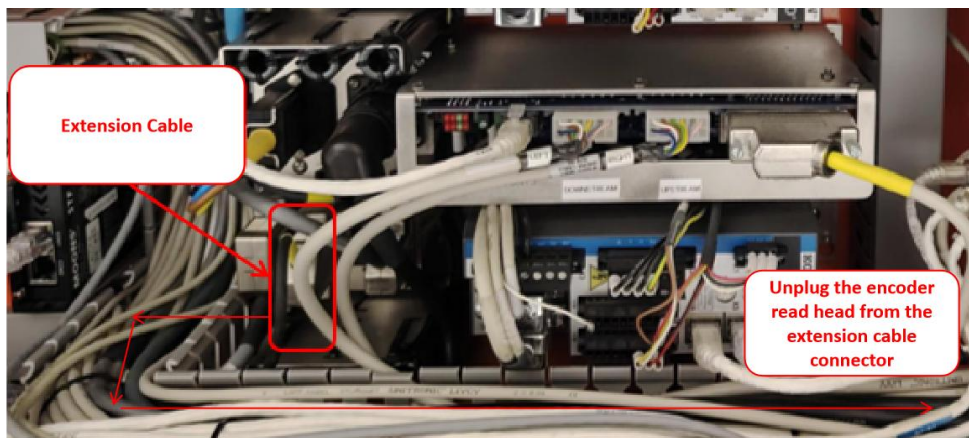


Figure 9:Y-Axis Kollmorgen Kit

	Work Instruction	Doc. Revision	00
	AXI V810 Y-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

12. **Removal of cable tie and the Y-Axis Encoder Bracket:** Remove the cable tie. Then, unfasten 2*M5x20 screws to remove the Y-Axis Read Head Encoder bracket. Take caution when removing the bracket to prevent damage to the read head encoder.

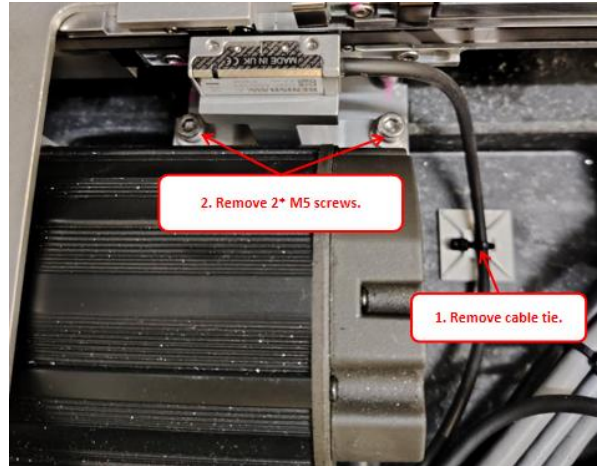


Figure 10:Y-Axis Read Head Encoder

13. **Dismantle the Read Head Encoder from the Bracket:** Unfasten the 2*M3x20 screws to remove the Read Head Encoder from the bracket.

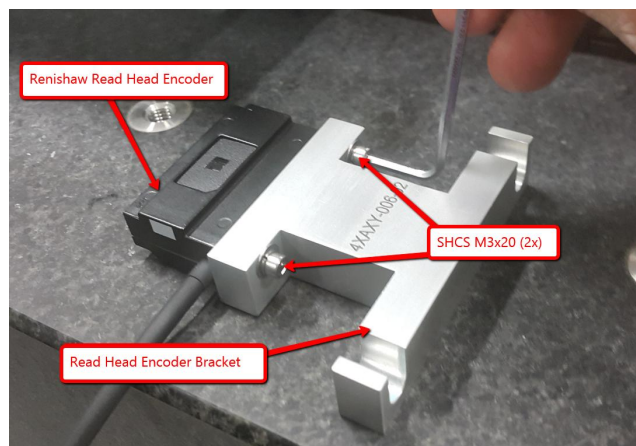


Figure 11:Y-Axis Read Head Encoder with bracket (Sample Image)

	Work Instruction	Doc. Revision	00
	AXI V810 Y-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

14. **Removal of Y-Axis Read Head Encoder:** Remove Y-Axis Read Head Encoder by referring to the routing below.

Note: Follow the red arrow for removal direction, and remove any necessary cable ties.

a) For S2EX and XLT

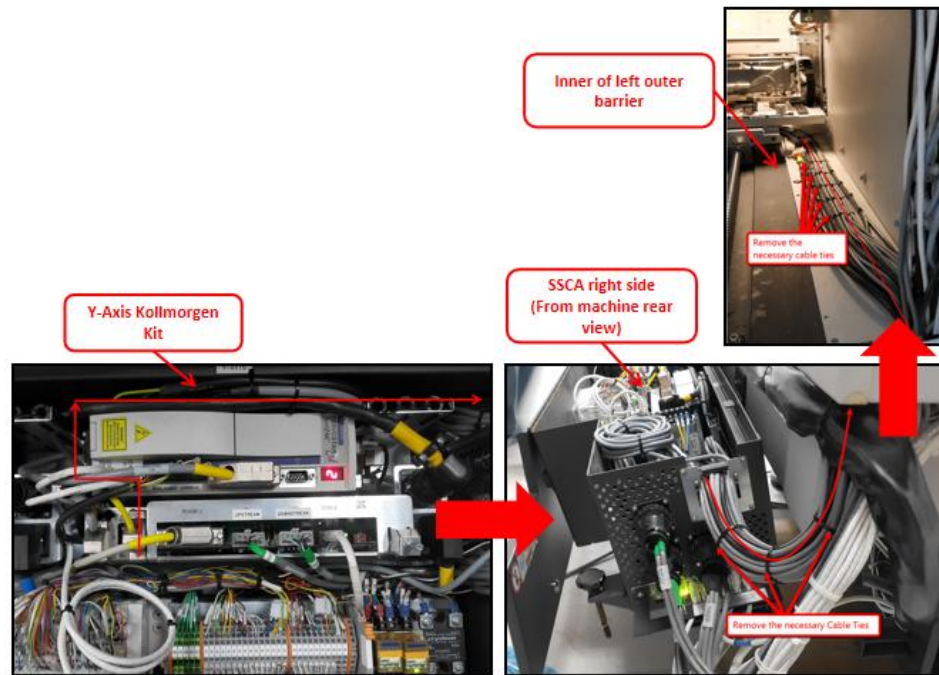


Figure 12:Y-Axis Read Head Encoder cable routing (Sample Image)

b) For S3

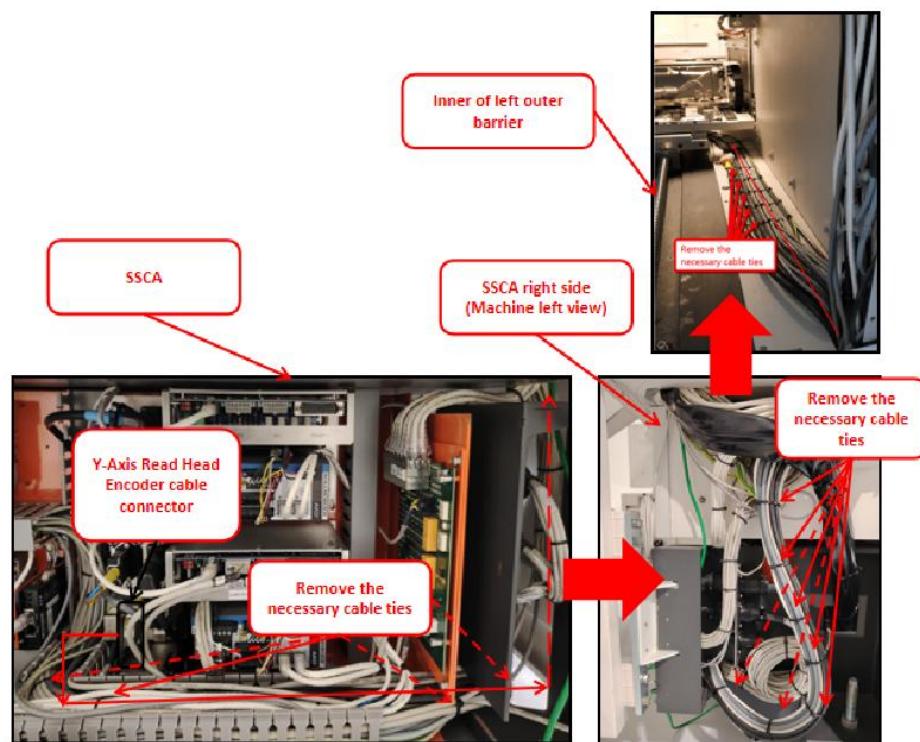


Figure 13:Y-Axis Read Head Encoder cable routing

	Work Instruction	Doc. Revision	00
	AXI V810 Y-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

New Y-Axis Read Head Encoder Replacement

15. **Spare Part Preparation:** Refer to the table below to gather and prepare the new spare part required for the replacement process.

Part number	Part Description	QTY	UOM	illustration
2675-0053	Encoder Head, Optical, 0.5UM	1	PC	

Table 2: Spare Part 2675-0053

16. **Read Head Encoder and Bracket Assembly:** Assemble new Y-Axis Read Head Encoder to the bracket by using 2*SHCS M3x20 and tighten the screws.

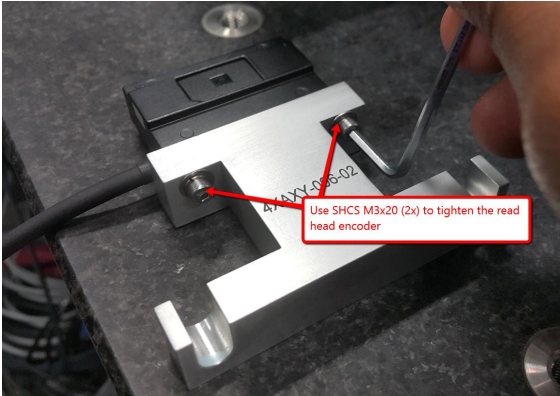


Figure 14:Y-Axis Read Head Encoder with bracket (Sample Image)

17. **Read Head Encoder Bracket Assembly:** Install the Read Head Encoder bracket on the machine and tighten the 2*M5x20 screws. Then, apply a cable tie to avoid cable damage during motion.

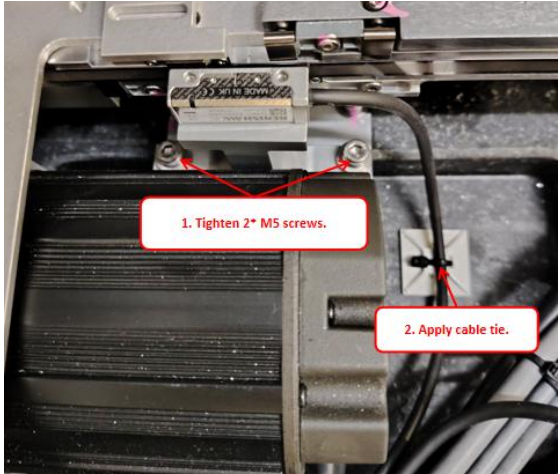


Figure 15:Y-Axis Read Head Encoder Bracket with Encoder Head

	Work Instruction	Doc. Revision	00
	AXI V810 Y-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

18. **Cable Routing:** Route the cable following the impacted area numbering in sequence.

Note: the cable should route thru underneath of servo motor.

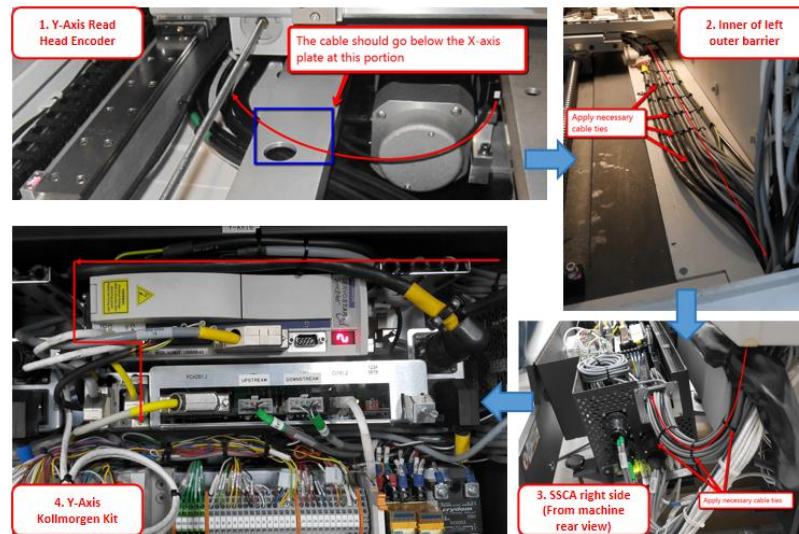


Figure 16:Y-Axis Read Head Encoder Cable Routing

19. **Re-connect the Y-Axis Read Head Encoder Connector:** Plug in the Y-Axis Read Head Encoder connector in the Y-Axis Kollmorgen kit.

a) For S2EX and XLT

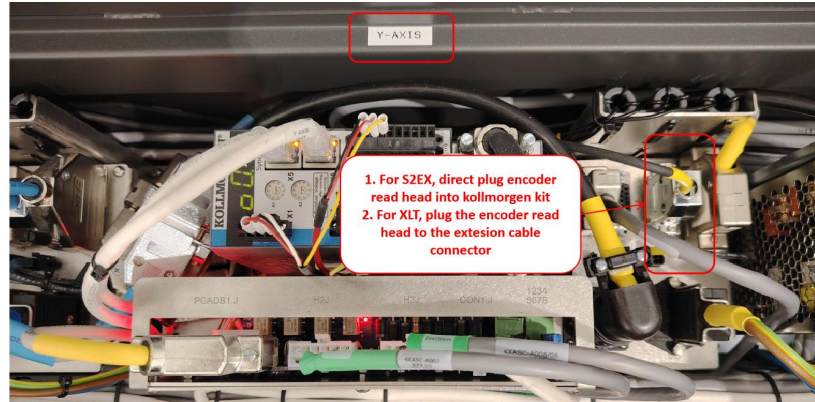


Figure 17:Y-Axis Kollmorgen Kit

b) For S3

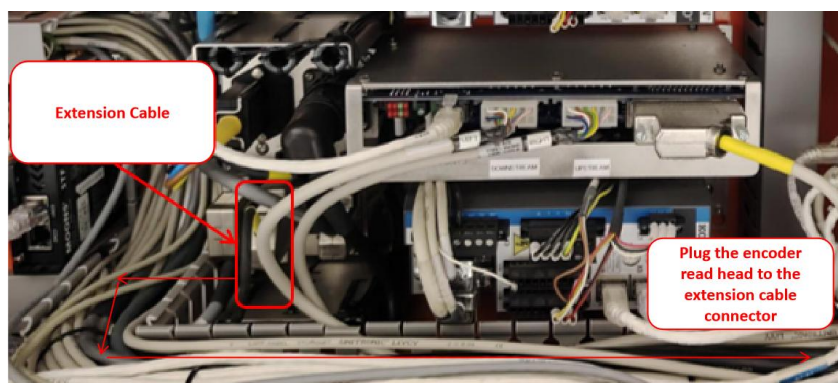


Figure 18:Y-Axis Kollmorgen Kit

	Work Instruction	Doc. Revision	00
	AXI V810 Y-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

20. **Cable Clamp Assembly:** Place the cable clamp on its position and tighten back the 2 screws.

a) For S2EX and XLT

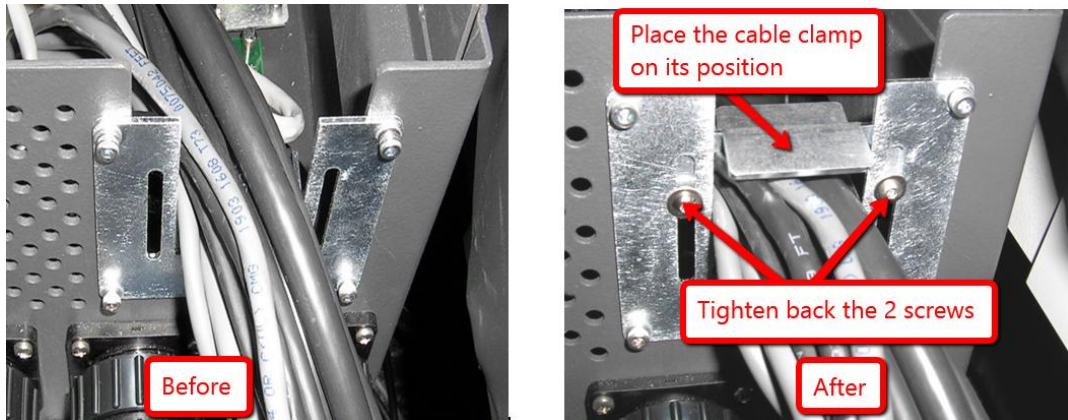


Figure 19:SSCA (Sample Image)

b) For S3

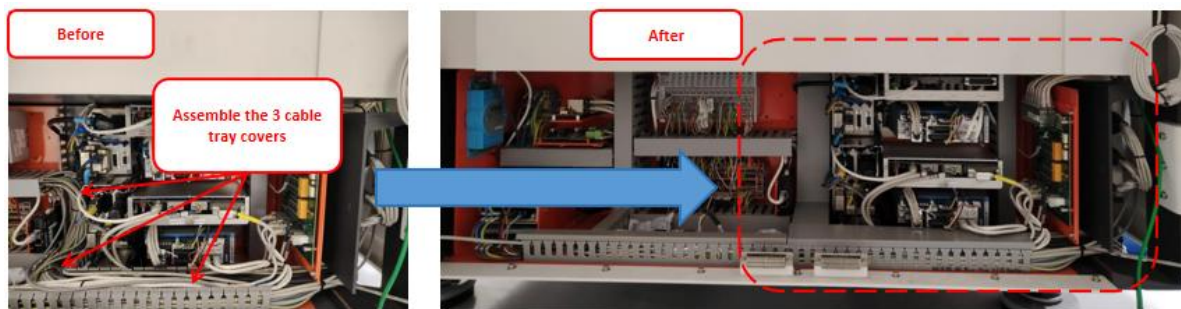


Figure 20:SSCA

21. **Horizontal Lead Cover Assembly:** Place back the horizontal lead cover and push it in the direction as shown in below figure. Then, tighten back the 3*M5 screws.

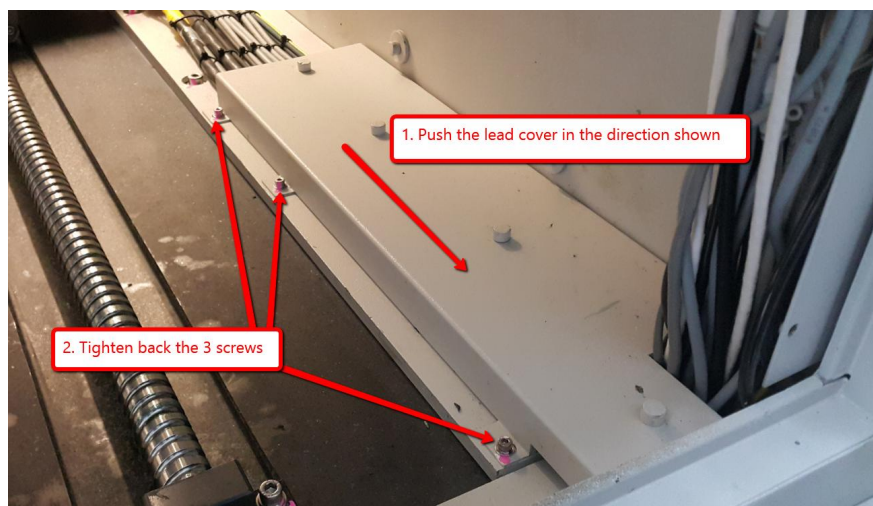


Figure 21:Horizontal Lead Cover (Sample Image)

	Work Instruction	Doc. Revision	00
	AXI V810 Y-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

22. **Vertical Lead Cover Assembly:** Assemble the vertical lead cover and tighten the screws.

a) For S2EX and XLT

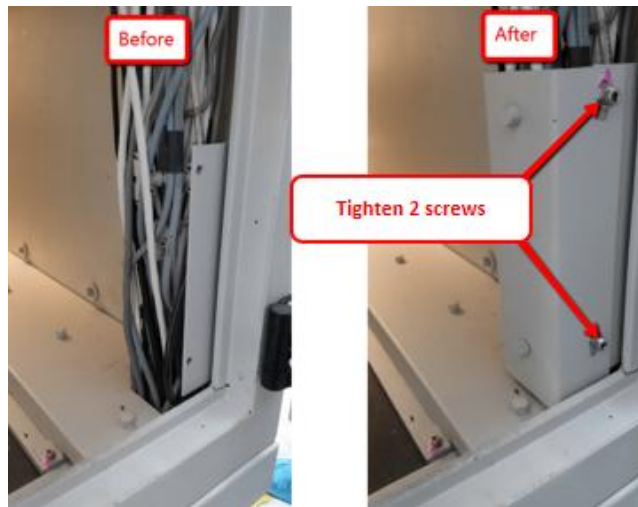


Figure 22:Vertical Lead Cover (Sample Image)

b) For S3

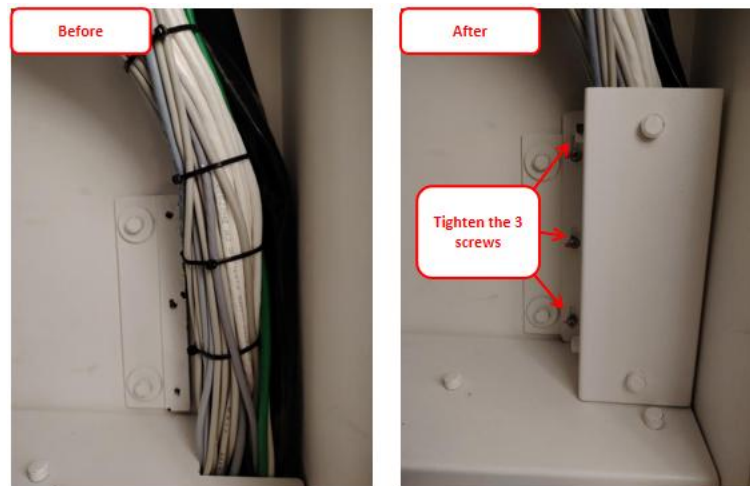


Figure 23:Vertical Lead Cover

	Work Instruction	Doc. Revision	00
	AXI V810 Y-Axis Quantic Encoder Read Head Removal and Replacement Procedure	Effective Date	30 Apr 2024

Tool Removal and Cleaning

23. Before concluding the replacement procedure, carefully inspect the machine interior to ensure that all tools, equipment, and debris have been removed. Retrieve any tools or equipment left inside the machine and return them to their designated storage area to prevent any interference with machine operation.
24. Dispose of any removed cable ties, packaging materials, or other debris in the designated waste disposal containers according to company guidelines or environmental regulations.
25. Once the interior surfaces have been cleaned and cleared of debris, visually inspect the machine to ensure that it is in a tidy and organized state, ready for operation.
26. Perform a final check to verify that all components have been properly installed and secured according to manufacturer specifications. Confirm that all cables are properly routed and secured within their respective E-chains to prevent any interference with machine movement.
27. Once the machine has been cleaned and inspected, power on the system according to standard operating procedures but without launch the V810 GUI.

Reference for Alignment Procedure

28. **Y-Axis Encoder Read Head Alignment:** Refer to “AXI V810 Quantic Encoder Read Head Alignment Procedure” to verify the encoder status and perform necessary alignment if needed.
Note: Crucial to ensure the signal strength for encoder read head along the y-axis stroke.